

Section B

Complete Class Note

Set-wise exam preparation • Bangla explanation • verified formulas • clean sketches • matching question-bank map

Set B1

Force System • Equilibrium • FBD • Supports • Resultant • Wheel

Set B2

Shear Force & Bending Moment • Loads • Pulley

Set B3

Truss • Method of Joints/Sections • Plane Motion

Set B4

Friction • Belt • Wedge • Flexible Cord

Exam rule: 4 sets থেকে যেকোনো 3 set answer করতে হবে। তাই 3টি set খুব শক্তভাবে + 4th set backup হিসেবে প্রস্তুত রাখা সবচেয়ে নিরাপদ।

Made by 2223033

Prepared from supplied class notes,
set-wise notes & regular question papers

Study Map & Exam Strategy

এই নোটটি আপনার দেওয়া **Section B set-wise system** অনুযায়ী সাজানো। প্রশ্নপত্রে topic placement কিছু বছরে crossover হতে পারে; এখানে topic-family অনুসারে একই set-এ রাখা হয়েছে।

4

sets in Section B

3

sets must be answered

1

backup set recommended

Set	Core topics	What to master
B1	Force system, FBD, equilibrium, supports	Definitions + reactions + resultant + wheel-over-block / smooth-contact logic
B2	SFBM, loading, pulley	Equivalent loads + support reactions + section cut + ideal-rope tension tracing
B3	Truss, plane motion	Joints/sections + T/C sign + projectile trajectory
B4	Friction, belt, wedge, flexible cord	Impending motion + slide/tip + belt proof + cable parabola/catenary

3-out-of-4 strategy: B1 ও B2-তে recurring theory/procedure খুব বেশি। B4-এ friction/cable family বারবার আসে। B3 truss scoring হলেও diagram ও calculation তুলনামূলক সময় নেয়। তাই practice অনুযায়ী strongest 3 বেছে নিন, কিন্তু 4th set একেবারে বাদ দেবেন না।

2020 exception: আপনার নির্দেশ অনুযায়ী 2020-কে normal 4-set pattern-এ force করা হয়নি। এই নোটে 2020-এর নির্দিষ্ট question wording অনুমান করে যোগ করা হয়নি।

Notation policy

সব formula-তে proper mathematical notation ব্যবহার করা হয়েছে: ΣF_x , ΣF_y , ΣM , x^2 , T_1/T_2 , $e^{\mu\theta}$ । Angle measure না বলা থাকলে component নেওয়ার আগে angle কোন axis-এর সাথে তা দেখুন।

Golden routine: Sketch/FBD → known directions → components → equilibrium equation → unit check → physical sense check.

Force System, Equilibrium & FBD

Core idea: Statics-এর সবচেয়ে বড় skill হলো **ঠিক FBD আঁকা**। Equation ভুলের চেয়ে FBD ভুল বেশি বিপজ্জনক, কারণ ভুল force direction দিলে পরের সব calculation ভুল হবে।

1. Free Body Diagram (FBD)

Definition: কোনো body বা body-group-কে surrounding থেকে আলাদা করে তার উপর acting সব **external force, weight, applied force, support reaction, friction এবং external moment** দেখানো sketch-ই Free Body Diagram.

বাংলায়: বাস্তব structure-টাকে “একলা” করে ভাবুন। যেসব সংযোগ কেটে দিলেন, সেই সংযোগগুলোর জায়গায় তাদের reaction বসান।

FBD drawing checklist

1. কোন object/member isolate করবেন ঠিক করুন।
2. Support/rope/contact কেটে reaction/tension বসান।
3. Weight বাদ দিতে বলা না হলে weight দেখান।
4. Applied load/couple দেখান।
5. Known two-force member হলে force member-axis বরাবর দিন।
6. Axes ও positive direction লিখুন।

2. Transmissibility of Force

Rigid body-তে একটি force-কে তার **same line of action** বরাবর অন্য point-এ সরালেও বাহ্যিক equilibrium/motion effect অপরিবর্তিত থাকে, যদি magnitude ও direction একই থাকে।

Same magnitude + same direction + same line of action

VERY HIGH Matching Question Bank

2015 2016 2017 2018 2019 2021 2023 2024 2025

- Define FBD and draw FBD of different members.
- Define transmissibility, two-force member and equilibrium.
- 2024/2025 family: definitions combined with FBD of a compound structure.

Two-Force Member, Equilibrium & Supports

3. Two-Force Member

যে member-এর উপর কেবল দুইটি external force কাজ করে এবং member equilibrium-এ থাকে, force দুটো অবশ্যই **equal, opposite ও collinear** হবে।

Tension member

দুই প্রান্তে force বাইরে দিকে টানে। Cable/rope সাধারণত tension-only.

Compression member

দুই প্রান্তে force ভিতরের দিকে ঠেলে। Weightless boom/strut প্রায়ই compression member.

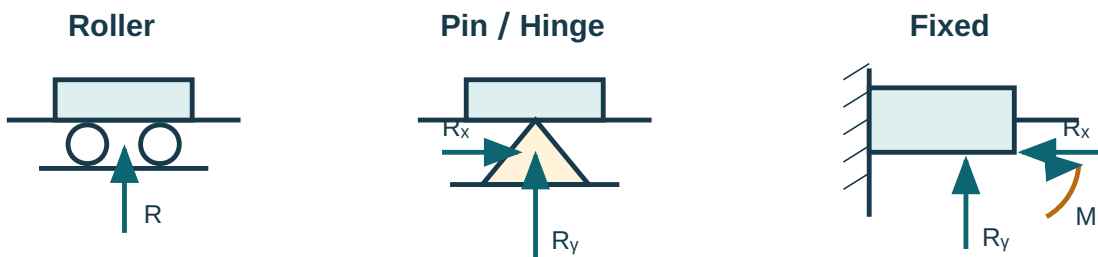
4. Equilibrium

Plane rigid body equilibrium-এর verified conditions:

$$\Sigma F_x = 0 \quad \Sigma F_y = 0 \quad \Sigma M_O = 0$$

এগুলো satisfied হলে resultant force ও resultant moment দুটোই zero.

5. Support Reactions



Support	Unknown reactions in 2D	What it allows	Memory trick
Roller	1 reaction, surface-এর normal	surface বরাবর movement + rotation	Roller = one arrow
Pin/Hinge	R_x, R_y	rotation allowed	Pin = two arrows
Fixed	R_x, R_y, M	translation/rotation restrained	Fixed = two arrows + moment

Common error: Roller reaction সবসময় vertical নয়। Inclined smooth surface হলে reaction surface-এর **normal** হবে।

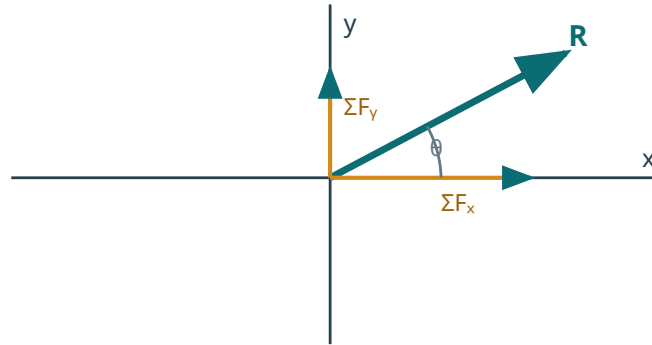
HIGH Matching Question Bank

2018 2019 2023 2024 2025

- Illustrate support systems and determine contact reactions.
- Smooth-contact sphere/cylinder problems: reaction normal to each smooth surface.
- Draw FBD of individual members after replacing supports by reactions.

Resultant Force, Moment, Couple & Torque

6. Resultant of Coplanar Concurrent Forces



$$R_x = \Sigma F_x, \quad R_y = \Sigma F_y, \quad R = \sqrt{R_x^2 + R_y^2}$$

$$\theta = \text{atan2}(R_y, R_x) \rightarrow \text{quadrant automatic}$$

Shortcut: diagram-এ force কোন quadrant-এ আছে আগে $\pm$ sign ঠিক করুন। Angle x-axis থেকে হলে $F_x = F \cos\theta$, $F_y = F \sin\theta$. y-axis থেকে হলে বিপরীত হতে পারে - blindly মুখস্থ করবেন না।

7. Moment of a Force

$$M_O = F \times d_{\perp}$$

এখানে $d_{\perp}$ হলো point O থেকে force-এর line of action পর্যন্ত **perpendicular distance**.

8. Couple and Torque

Term	Meaning	Formula
Couple	equal & opposite parallel forces, different lines of action	$M = F s$
Torque	force-এর rotational effect about an axis	$\tau = r \times F$

Moment shortcut: Force component ভাঙার আগে দেখুন সরাসরি perpendicular distance পাওয়া যায় কি না। সহজ axis নিলে equation ছোট হয়।

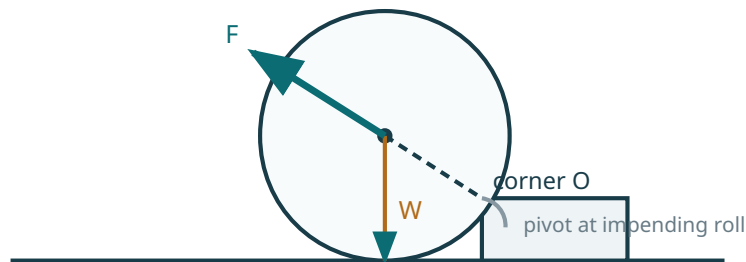
MEDIUM-HIGH Matching Question Bank

2016 2019 2023

- Find resultant force and its angle from the x-axis.
- Define/identify moment, couple and torque.
- Bell-crank/contact-force type equilibrium problems.

Wheel Over a Block & Smooth Contact

9. Wheel just starts rolling over a corner



Impending rolling-over অবস্থায় wheel ground থেকে lift করতে শুরু করে। তখন ground reaction **zero**, এবং block corner O instantaneous pivot হিসেবে কাজ করে।

Safe procedure

1. Geometry থেকে center C, corner O এবং relevant angles বের করুন।
2. FBD-তে W, applied F এবং corner reaction R দেখান।
3. **Moment about O** নিন - R eliminate হবে।
4. Least force চাইলে F-এর moment arm maximum হওয়া দরকার; তাই force direction optimize করুন।

Do not memorize one numeric angle. Block height/radius বদলালে geometry বদলে যায়।

10. Smooth Contact Rule

Smooth contact $\rightarrow$ friction = 0 $\rightarrow$ reaction is normal to the contact surface

Sphere against wall: wall reaction horizontal; inclined plane reaction plane-এর normal.

Rope on smooth body: tension rope direction বরাবর; contact reaction normal.

HIGH Matching Question Bank

2015 2016 2017 2021 2023 2024 2025

- Wheel on point of rolling over a block: find least force and reaction.
- Smooth plane/wall sphere or cylinder: determine reactions.
- 2025 cylinder on smooth incline with rope: find plane inclination and reaction.

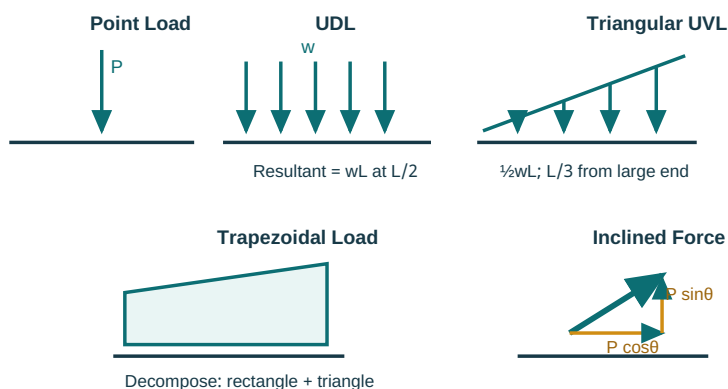
Shear Force & Bending Moment

1. Definitions

Quantity	Exam-ready definition	Unit
Shear Force, V	At a section, one side of the cut-এ section-এর parallel external forces-এর algebraic sum.	N, kN, lb, kip
Bending Moment, M	At a section, one side of the cut-এ সব external force-এর moments-এর algebraic sum.	N·m, kN·m, lb-ft

বাংলায়: Beam কাটলে ভিতরে দুইটি action প্রকাশ পায় - vertical balance রাখে V , rotation balance রাখে M .

2. Loading Types & Equivalent Resultants



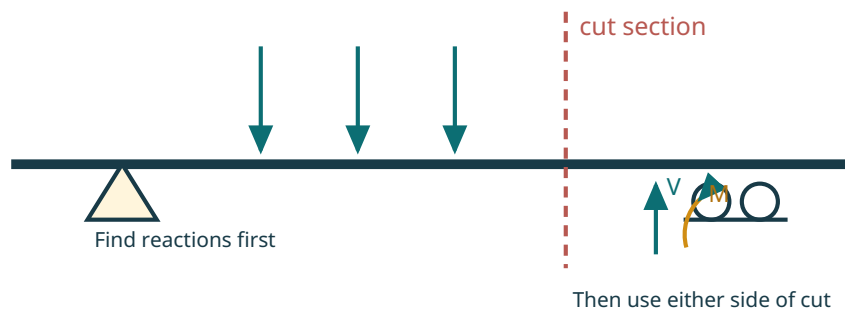
Load	Equivalent resultant	Location
Point P	P	same point
UDL w over L	wL	midpoint, $L/2$
Triangular UVL $0 \rightarrow w$	$\frac{1}{2}wL$	$L/3$ from larger-intensity end
Trapezoidal $w_1 \rightarrow w_2$	$\frac{1}{2}(w_1 + w_2)L$	best: rectangle + triangle decomposition

VERY HIGH Matching Question Bank

2015 2016 2017 2018 2019 2021 2023 2024 2025

- Define shear force and bending moment.
- Find support reactions and V , M at specified sections of a loaded beam.

SFBM - Step-by-Step Solving Method



3. Procedure

1. **Full-beam FBD:** supports replace করে reactions বসান।
2. Distributed loadগুলো equivalent point load-এ convert করুন।
3. $\sum F_x=0$, $\sum F_y=0$, $\sum M=0$ দিয়ে support reactions বের করুন।
4. Question-এর section-এ beam cut করুন।
5. Left অথবা right side নিন - যেকোনো load কম, সেটি usually easier.
6. Cut face-এ unknown V ও M assume করুন, equilibrium solve করুন।
7. অন্য side দিয়ে check করলে magnitude একই আসা উচিত (sign convention consistent রাখতে হবে)।

Fast exam technique: প্রথমে support-এর moment নিলে ঐ support reaction equation থেকে disappear করে। তারপর vertical equilibrium দিয়ে অন্য reaction.

4. Sign Convention - practical rule

Class notes-এ external moment-এর জন্য clockwise-positive convention অনেক example-এ ব্যবহার করা হয়েছে। Internal V/M-এর ক্ষেত্রে একটি **convention** শুরুতে **declare করে শেষ পর্যন্ত consistent** থাকুন। Answer-এর physical direction/diagram অবশ্যই দেখান।

Dimension check: Shear is force; bending moment is force $\times$ length. If you obtain kN for M, unit missing.

5. Sanity checks

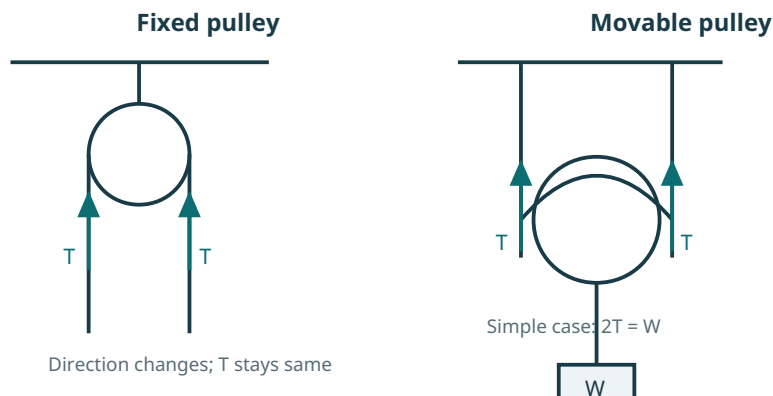
- Point couple shear force-এ সরাসরি যোগ হয় না; moment equation-এ যোগ হয়।
- UDL resultant wL এবং centroid ঠিক জায়গায় না বসালে reaction ভুল হবে।
- Inclined load-এর component angle-reference দেখে নিন।

VERY HIGH Matching Question Bank

2018 2019 2023 2024 2025

- Large mixed-load beam with UDL/UVL, concentrated load and applied couple.
- Find V and M at (a)-(a), (b)-(b) sections; show reactions.

Pulley Systems - Verified Method



6. Ideal pulley assumptions

- Frictionless + massless pulley.
- **Same continuous rope** → **same tension T everywhere on that rope.**
- Fixed pulley changes direction only.
- A movable block supported by n segments has upward support nT, if those n segments belong to the same ideal rope and no extra forces act.

Important correction to shortcut: W/2, W/4, W/8 patterns are not universal. They work only after you identify how many rope segments support each moving block. Different ropes can have different tensions.

7. Cascaded pulley method

1. Start from the **lowest moving pulley/load**.
2. Write equilibrium of that pulley block.
3. Find its rope tension.
4. Move one level upward and repeat.
5. Finally transfer the rope force to the beam and use beam moment equilibrium.

বাংলা memory trick: “নিচ থেকে tension তুলতে তুলতে উপরে উঠি।” Fixed pulley-কে weight-divider ভাববেন না; এটি rope direction ঘোঁরায়।

8. Useful class-note answer patterns

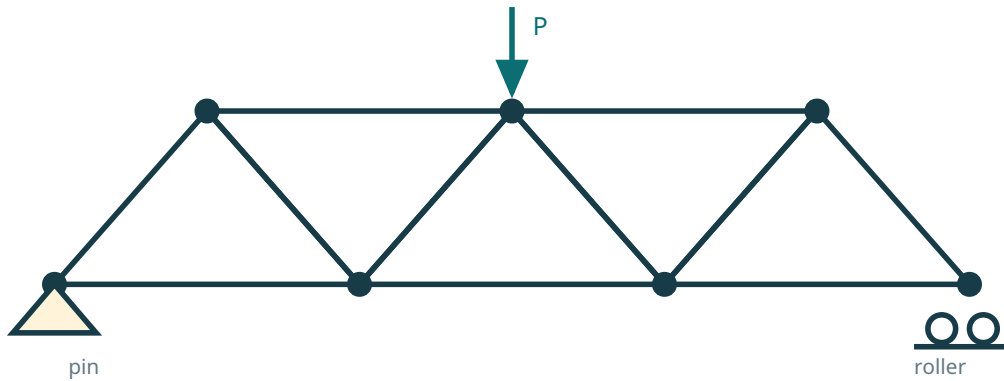
Pattern	Source-note result	Purpose
Beam + compound pulley + 700 lb + UDL	$W = 21,500 \text{ lb}$	tension propagation + beam moment
Compound pulley repeated QB	$W = 5,400 \text{ lb}$	multi-stage tension
Another chain	$W = 4,800 \text{ lb}$	$W/2 \rightarrow W/4 \rightarrow W/8$ tracing
Reaction at A problem (2019)	$V_A = -25 \text{ lb}$	negative reaction means actual direction downward

HIGH Matching Question Bank

2016 2017 2018 2019 2023 2024 2025

- Find W for equilibrium, assuming frictionless weightless pulleys.
- Determine reaction at a beam support connected to a pulley system.

Truss Fundamentals



1. What is a truss?

A truss is a framework of straight members connected to form a stable triangulated system. In ideal analysis, joints are pin-connected and loads act at joints.

বাংলায়: Triangle geometry-এর কারণে member-গুলো মূলত axial tension/compression নেয়; bending neglect করা হয় ideal truss model-এ।

2. Fundamental assumptions

- Joints ideal frictionless pins/hinges.
- Loads/reactions applied at joints.
- Members are straight two-force members.
- Member self-weight neglected or converted to joint loads if necessary.
- No member bending in ideal analysis.

3. Common truss types

Type	Visual memory	Common use
Pratt	diagonals usually lean toward span center	bridge/roof
Howe	diagonals opposite Pratt orientation	roof/bridge
Warren	repeating triangles	bridge
Fink	fan-like roof web	roof

4. Tension / Compression sign

At a joint, initially assume unknown member force **away from joint** = tension. Result negative হলে member actually **compression**.

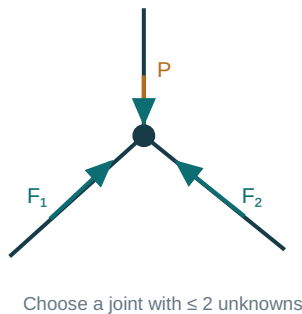
VERY HIGH Matching Question Bank

2015 2016 2017 2018 2019 2021 2023 2024 2025

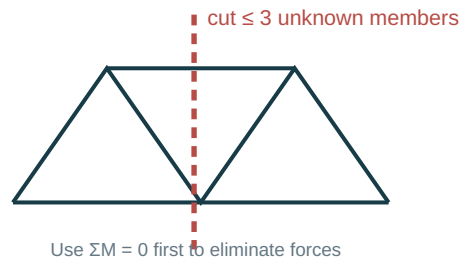
- Define truss and state fundamental assumptions.
- Draw/identify types of roof/bridge trusses.
- Determine external reactions and member forces; state T or C.

Method of Joints & Method of Sections

Method of Joints



Method of Sections



5. Method of Joints

1. Whole truss-এর reactions বের করুন.
2. যে joint-এ maximum 2 unknown member force, সেই joint দিয়ে শুরু করুন.
3. Unknown arrows tension ধরে joint থেকে বাইরে আঁকুন.
4. $\Sigma F_x = 0$ এবং $\Sigma F_y = 0$ solve করুন.
5. Negative $\rightarrow$ compression; positive $\rightarrow$ assumed tension correct.

6. Method of Sections

1. Target member-গুলোর মধ্য দিয়ে একটি imaginary cut দিন; ideally ≤ 3 unknown members কাটুন.
2. এক পাশ isolate করুন.
3. যে point-এ দুই unknown force intersect করে, সেই point-এর moment নিলে দুটো eliminate হয়.
4. Then ΣF equations.

Fast method-of-section rule: "Moment first, components later." Target member-এর force দ্রুত বের করতে intersection point smartly choose করুন.

7. Zero-force member rules

Joint condition	Conclusion
Unloaded joint; only 2 non-collinear members	Both are zero-force members.
Unloaded joint; 3 members; 2 collinear	The third non-collinear member is zero.

Zero-force member মানে member useless নয়; অন্য loading case-এ force নিতে পারে এবং structure stability-তে প্রয়োজন হতে পারে.

VERY HIGH Matching Question Bank

2016 2017 2019 2023 2024 2025

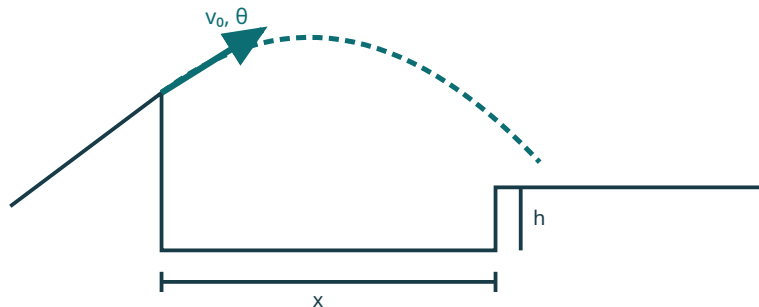
- Use method of sections to determine specified members.
- Find all member forces and external reactions.
- 2024 includes zero-force member in theory family.

Plane Motion & Projectile-Type Problems

8. Plane motion

A rigid body has plane motion when its motion can be described in a plane; points of the body move in planes parallel to one fixed plane. A special case is **rectilinear translation**, where points move along parallel straight lines.

বাংলা: 2D plane-এর ভিতরে motion analysis - x ও y দুই direction-এ position/velocity দেখা হয়। Motorcycle ditch problem-এ rider/body-কে particle ধরে projectile motion use করা হয়।



9. Projectile equations

$$x = v_0 \cos\theta \cdot t$$

$$y = v_0 \sin\theta \cdot t - \frac{1}{2}gt^2$$

$$y = x \tan\theta - (g x^2)/(2v_0^2 \cos^2\theta)$$

10. Motorcycle-ditch technique

1. Units unify করুন (mph → ft/s or m/s).
2. Horizontal gap x known; trajectory equation directly use করলে t eliminate হয়.
3. Landing-side elevation sign carefully define করুন.
4. Question “maximum h ” হলে launch level থেকে projectile y + given ground-level offset combine করুন.

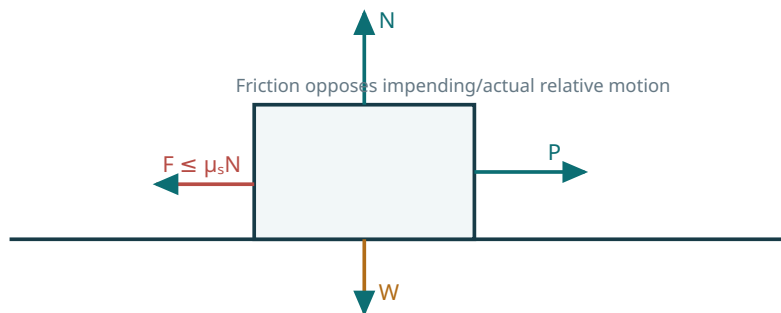
Source variants differ: different years use different speeds/geometry. তাই 20 ft/10 ft or a specific final number মুখস্থ নয় - equation pattern মুখস্থ করুন.

MEDIUM Matching Question Bank

2015 2017 2018 2021 2025

- Motorcycle stunt rider jumps a ditch: determine maximum h .
- Define plane/rectilinear motion in a mixed theory question.

Friction Fundamentals



1. Friction

Friction is the tangential resisting force between contacting surfaces that opposes **relative motion or impending relative motion**.

State	Relation	Meaning
Static	$0 \leq F_s \leq \mu_s N$	force adjusts as needed up to limiting value
Impending motion	$F_{s,max} = \mu_s N$	just about to move
Kinetic	$F_k = \mu_k N$	surfaces already sliding

বাংলা: Static friction সবসময় μN নয়। কেবল **limiting/impending** অবস্থায় $F = \mu N$.

2. Angle of friction & angle of repose

Angle of friction, φ : resultant contact reaction R এবং normal N -এর মধ্যকার angle at limiting friction.

$$\tan \varphi = \mu$$

Angle of repose, α : inclined plane-এর সেই angle যেখানে body gravity-এর কারণে just slide করতে যাচ্ছে. Coulomb model-এ:

$$\alpha = \varphi \quad \text{and} \quad \tan \alpha = \mu$$

3. Basic laws used in this course

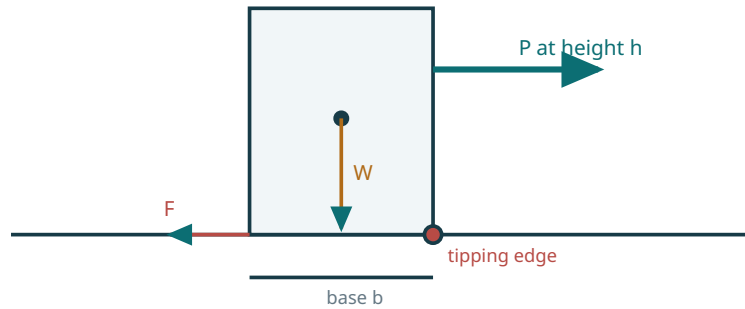
- Limiting friction proportional to normal reaction.
- Basic dry-friction model-এ coefficient apparent contact area-এর উপর নির্ভর করে না.
- Direction always opposes actual/impending relative motion.

VERY HIGH Matching Question Bank

2015 2016 2017 2018 2019 2021 2023 2024 2025

- Distinguish angle of friction and angle of repose.
- State laws of friction.
- Two-body friction systems with pulleys: find T and unknown weight.

Sliding vs Tipping / Overturning



4. Compare two thresholds

Gradually increasing horizontal force P applied at height h :

Sliding threshold:

$$P_{\text{slide}} = \mu_s W$$

(level floor, no vertical P component)

Tipping threshold:

$$P_{\text{tip}} \cdot h = W \cdot (b/2)$$

$$P_{\text{tip}} = Wb/(2h)$$

Comparison	What happens first
$P_{\text{tip}} < P_{\text{slide}}$	Tips before sliding
$P_{\text{slide}} < P_{\text{tip}}$	Slides before tipping

Class-note shortcut: দুইটা আলাদা “required force” বের করুন। যেটা smaller, সেই failure mode আগে হবে।

5. Inclined plane version

Inclined plane-এ weight components, normal reaction এবং friction আগে resolve করুন। Tipping-এর সময় resultant contact reaction limiting edge-এর মধ্য দিয়ে যায়।

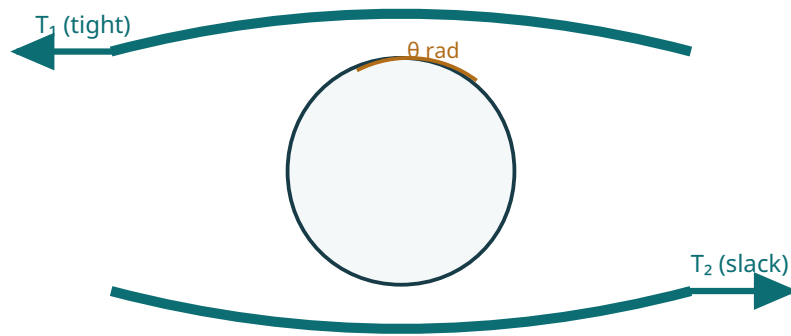
At tipping, contact pressure no longer acts at the center of base. Moment about the tipping edge is the cleanest equation.

VERY HIGH Matching Question Bank

2015 2016 2017 2018 2021 2024 2025

- Tall/slender body: will it slide or tip over as Q increases?
- Find largest application height so that sliding occurs before tipping.
- 2025: body on a 30° incline, determine slip or overturn.

Belt Friction



6. Tight-side and slack-side tension

For a belt just on the point of slipping over a rough cylinder/pulley:

$$T_1 / T_2 = e^{\mu\theta}$$

where T_1 = tight-side tension, T_2 = slack-side tension, μ = coefficient of friction, θ = angle of contact **in radians**.

7. Proof skeleton - exam writing

1. Take a small belt element subtending $d\theta$.
2. Let tension rise from T to $T+dT$, normal reaction dN .
3. Radial equilibrium for very small $d\theta$ gives approximately $dN = T d\theta$.
4. At impending slip, $dF = \mu dN = \mu T d\theta$.
5. Tangential equilibrium gives $dT = \mu T d\theta$.
6. Integrate from T_2 to T_1 and 0 to θ :

$$\int (dT/T) = \mu \int d\theta \rightarrow \ln(T_1/T_2) = \mu\theta \rightarrow T_1 = T_2 e^{\mu\theta}$$

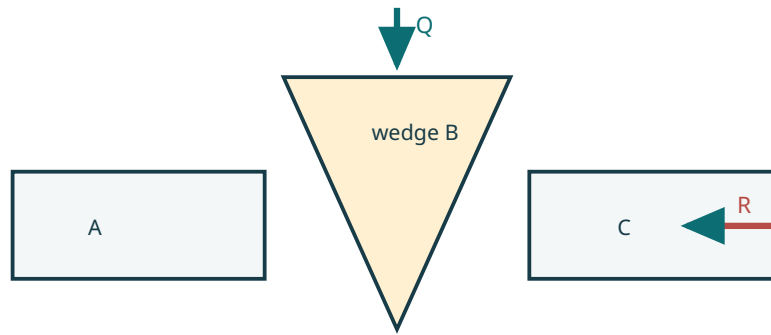
Common error: θ degrees-এ দিলে formula ভুল হবে. Degrees $\rightarrow$ radians: $\theta_{\text{rad}} = \theta^\circ \times \pi/180$.

HIGH Matching Question Bank

2015 2019 2021 2024

- For belt friction, prove $T_1 = T_2 e^{\mu\theta}$.
- Apply tight/slack tension relation in frictional pulley systems.

Wedge Friction



8. Wedge

A wedge is a solid tapered block, often V-shaped in cross-section, driven between bodies to lift, separate or produce a large normal force.

9. Friction-angle method

$$\varphi = \tan^{-1} \mu$$

At each rough contact, instead of drawing N and F separately, resultant reaction R can be drawn at angle φ from the normal, tilted **opposite the impending relative motion**.

Wedge problem workflow

1. Decide impending motion of each body.
2. Draw separate FBD of block C, wedge B, and other contacting body A if needed.
3. At each contact, place friction direction opposite relative impending motion.
4. Use $\varphi = \tan^{-1} \mu$ to combine $N+F$ into resultant R , if convenient.
5. Solve block first if its external R/W data are known; then wedge equilibrium.

বাংলা memory: “যেদিকে slip হতে চাইবে, friction তার উল্টো।” কিন্তু দুই body-র relative motion আলাদা হতে পারে - একই contact-এর action-reaction force direction opposite হবে.

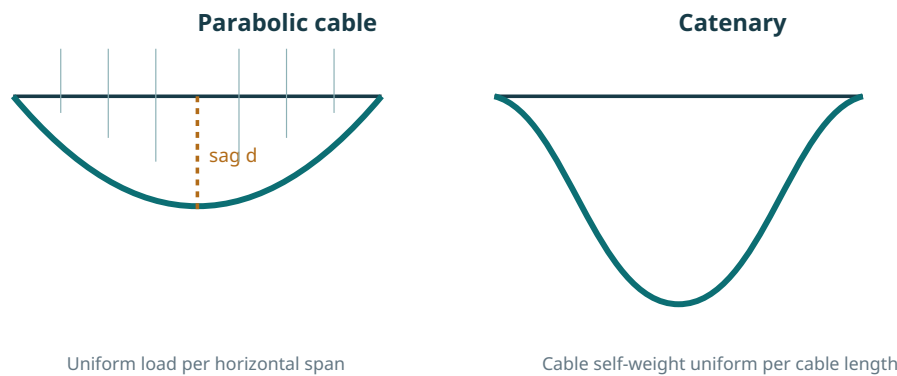
Wedge angle, friction angle এবং reaction angle গুলিয়ে ফেলবেন না. Diagram-এ প্রতিটি angle label করুন.

HIGH Matching Question Bank

2016 2017 2018 2019 2023 2024 2025

- Define wedge and determine force Q for impending motion of movable block C.
- Wedge with W_A , W_B , W_C , horizontal R and common coefficient of friction.

Flexible Cord: Parabola vs Catenary



Feature	Parabolic chord/cable	Catenary
Loading basis	uniform load per horizontal span	uniform self-weight per actual cable length
Typical shape	parabola	cosh curve
Course shortcut	light cable / small sag may be approximated as parabola	self-weight-dominated cable, especially larger sag
Equation	$y = wx^2/(2H)$	$y = K \cosh(x/K)$

Notation: Class note often uses **Q** for horizontal tension. এখানে clarity-এর জন্য H লিখছি; অর্থাৎ $H = Q$.

10. Parabolic cable derivation

Lowest point as origin; horizontal distributed load w per unit horizontal length. At a section x away:

$$H = T \cos\theta, \quad wx = T \sin\theta$$

$$\tan\theta = dy/dx = wx/H$$

$$\text{Integrate} \rightarrow y = wx^2/(2H)$$

Same-level supports, span L , sag d at $x=L/2$:

$$d = wL^2/(8H) \rightarrow H = wL^2/(8d)$$

HIGH Matching Question Bank

2015 2023 2025

- What is a parabolic chord? Derive $y = wx^2/(2Q)$.
- For equal-level supports prove $Q = wL^2/(8d)$.

Flexible Cord: Tension, Length & Catenary

11. Parabolic-cable tension

$$T(x) = \sqrt{(H^2 + (wx)^2)}$$

Same-level supports হলে maximum tension support-এ, $x=L/2$.

$$T_{\max} = \sqrt{(H^2 + (wL/2)^2)}$$

Small-sag parabolic approximate length:

$$S \approx L + 8d^2/(3L)$$

12. Catenary derivation roadmap

Self-weight per actual cable length = w . Let $K = H/w$. Lowest point origin; arc length from lowest point = s .

$$H = T \cos\theta, \quad ws = T \sin\theta$$

$$\tan\theta = dy/dx = s/K$$

Using $ds^2 = dx^2 + dy^2$ and integration:

$$s = K \sinh(x/K)$$

$$y = K \cosh(x/K)$$

$$T = w y = \sqrt{(H^2 + w^2 s^2)}$$

Memory trick: Catenary trio: $s \rightarrow \sinh$, $y \rightarrow \cosh$, $K=H/w$.

Cable question-এ “weight per ft of cable” আর “load per horizontal ft” এক জিনিস নয়. এটিই parabola बनाম catenary নির্বাচন করার মূল clue.

HIGH Matching Question Bank

2017 2023 2024 2025

- Cable suspended between same-level supports: find cable length, lowest-point tension and maximum tension.
- 60-ft cable weighing 360 lb, sag 15 ft: find support tension/span-type quantities.
- Different years vary sag/span; method remains the same.

Topic-to-Year Master Map

Observed from the supplied regular papers. 2020 is excluded from normal mapping by instruction. The map prioritizes recurring topic families; exact numerical data change year to year.

Topic family	Observed years	Priority	Typical question form
FBD / two-force / transmissibility / equilibrium	2015, 2016, 2017, 2018, 2019, 2021, 2023, 2024, 2025	A+	definitions + FBD + contact/support reaction
Beam SFBM	2015, 2016, 2017, 2018, 2019, 2021, 2023, 2024, 2025	A+	reactions + V/M at sections
Truss	2015, 2016, 2017, 2018, 2019, 2021, 2023, 2024, 2025	A+	specified members / all members, T or C
Friction / angle / slide-tip	2015, 2016, 2017, 2018, 2019, 2021, 2023, 2024, 2025	A+	theory + impending motion calculation
Pulley	2016, 2017, 2018, 2019, 2023, 2024, 2025	A	find W / reaction for equilibrium
Wedge	2016, 2017, 2018, 2019, 2023, 2024, 2025	A	find Q at impending motion
Flexible cord/cable	2015, 2017, 2021, 2023, 2024, 2025	A	parabola/catenary derivation + tension/sag
Belt friction proof	2015, 2019, 2021, 2024	B+	derive $T_1/T_2 = e^{\mu\theta}$
Plane / projectile motion	2015, 2017, 2018, 2021, 2025	B+	motorcycle ditch / rectilinear definition

Interpretation: The safest preparation is not “predict one exact numeric.” Prepare the **method pattern**, because dimensions/loads change while structure of the question repeats.

Theory & Derivation Checklist

Must-write theory

1. Free Body Diagram
2. Transmissibility of force
3. Two-force member
4. Equilibrium
5. Tension vs compression member
6. Support types & reactions
7. Shear force and bending moment
8. Moment, couple, torque
9. Truss + assumptions + types
10. Angle of friction vs repose
11. Laws of friction
12. Wedge
13. Parabolic chord vs catenary
14. Plane/rectilinear motion

High-value derivations

1. $T_1/T_2 = e^{\mu\theta}$ - belt friction
2. $y = wx^2/(2H)$ - parabolic cable
3. $H = wL^2/(8d)$ - same-level supports
4. Catenary: $s=K \sinh(x/K)$, $y=K \cosh(x/K)$
5. Load-resultant centroid rules (UDL/UVL)

Numerical patterns

- Wheel over block
- Mixed-load beam
- Compound pulley
- Truss joints/sections
- Slide vs tip
- Wedge friction
- Cable tension/sag/span
- Motorcycle trajectory

3-set answer plan: Before exam, complete one timed problem from each B1-B4. In exam, scan all four full sets first. Select the 3 where (i) diagram is familiar, (ii) data is clean, (iii) you can see the equation path within 60-90 seconds.

Do not skip units, FBD, or final direction. A negative reaction/member force is not automatically "wrong"; it usually means actual direction is opposite the assumed arrow.

Section B Formula Bank

Topic	Formula / Rule
Equilibrium	$\Sigma F_x=0, \Sigma F_y=0, \Sigma M=0$
Resultant	$R=\sqrt{[(\Sigma F_x)^2+(\Sigma F_y)^2]}$
Moment	$M=F d_{\perp}$
UDL	$W=wL$ at $L/2$
Triangular UVL	$W=\frac{1}{2}wL$, centroid= $L/3$ from large end
Movable pulley	$nT =$ supported load (only for identified supporting segments)
Truss joint	$\Sigma F_x=0, \Sigma F_y=0$
Projectile	$y=x \tan\theta - \frac{gx^2}{2v_0^2 \cos^2\theta}$
Static friction	$F_s \leq \mu_s N$; impending: $F=\mu_s N$
Friction angle	$\tan\phi=\mu$
Belt friction	$T_1/T_2=e^{\mu\theta}$, θ in rad
Tip threshold, level	$P_{\text{tip}}=Wb/(2h)$
Parabolic cable	$y=wx^2/(2H)$
Same-level sag	$H=wL^2/(8d)$
Parabolic cable tension	$T=\sqrt{H^2+(wx)^2}$
Small-sag cable length	$S\approx L+8d^2/(3L)$
Catenary	$K=H/w$; $s=K \sinh(x/K)$; $y=K \cosh(x/K)$; $T=wy$

Unit reminder: Force $\rightarrow$ N/kN/lb/kip. Moment $\rightarrow$ N·m/kN·m/lb·ft. μ dimensionless. Cable x, L, d একই length unit-এ রাখুন.

Last 10-minute memory line

FBD $\rightarrow$ Reaction $\rightarrow$ Resultant/Moment $\rightarrow$ Cut/Joint $\rightarrow$ Impending direction $\rightarrow$ Final sense check

Materials Used & Quality Notes

This compiled note is based on the supplied CE 1201 Section B set-wise notes, class notes and regular final examination papers. It follows the user-defined set distribution.

Category	Supplied material used
B1	Equilibrium/FBD note; Kabbo Sir moment & parallel coplanar forces; full class notes
B2	Shear Force & Bending Moment note; Pulley note
B3	Truss note; Plane Motion note
B4	Friction note; Flexible Cord note
QB matching	Regular final papers 2015-2021 bundle, 2023, 2024 and 2025

Verification approach: Class-note shortcuts were retained where useful, but universal rules were stated more carefully. Examples: pulley W/2-W/4 chains are treated as diagram-dependent; static friction is $F \leq \mu N$ until impending slip; roller reaction is normal to its contact surface.

Question-bank limitation: 2020 is an explicit exception per instruction and is not force-mapped. Numerical dimensions in past papers vary, so this note emphasizes transferable solution procedures.

CE 1201 - Engineering Mechanics

Section B Complete Class Note

Made by 2223033